

INDUSTRY PROJECT PROPOSAL

A. PROJECT IDENTIFICATION

A1. Project Title

Tea Collection Management System

A2. Project Track (tick one)



UCSC Track (UT)

Industry / open-source topic via
UGVLE



Industry Track (IT)

PDC-registered company project



Startup Track (ST)

Student-founded registered startup

A3. Project Type (tick one)



Software Product

Web / mobile / desktop application



Proof of Concept (PoC)

Research-based prototype / algorithm

B. GROUP & SUPERVISORY DETAILS

B1. Student Group

#	Full Name	Registration No.	Degree	Email
1	W. D. R. Migara	2023/CS/111	☒ CS ☐ IS	radeeshmigara43@gmail.com
2	U.H.N.Rodrigo	2023/CS/162	☒ CS ☐ IS	hasindunimantha2003@gmail.com
3	G.S.Naayanan	2023/IS/064	☐ CS ☒ IS	g.santhosh.n0702@gmail.com
4	R.E.M.K.R.B Ekanayake	2023/CS/042	☒ CS ☐ IS	kalanaekanayake2021@gmail.com
5	S.R.A.Patabedi	2023/CS/134	☒ CS ☐ IS	sandaruruwaneka21@gmail.com

B2. Academic Co-Supervisor (UCSC)

Name: Mr. Janitha

B3. Industry Mentor / Company

Name: Sachith Kasthuriarachchi

Organization: WSO2

C. PROJECT OVERVIEW

C1. Problem Statement

The tea industry still relies heavily on manual records and phone-based coordination. This causes inaccurate record keeping, communication delays, weight disputes, limited transparency, and operational inefficiencies among tea estate owners, managers, collection agents, receiving officers, and factories.

C2. Proposed Solution & Objectives

Develop a centralized Tea Collection Management System.

Objectives:

1. Digitize tea collection and delivery workflows.
2. Reduce paperwork and manual errors.
3. Improve transparency through OTP-based verification.
4. Manage fertilizer requests, approvals, and deductions.
5. Provide dashboards, reports, and analytics.

C3. Key Features / Modules

#	Feature / Module	Brief Description	Primary ILO(s)
1	Employee & Tea Plucking Management	Employee registration, manager assignment, and plucking records.	ILO-02
2	Tea Collection & Delivery Management	Selling requests, collection assignments, delivery tracking.	ILO-02
3	OTP Verification & Complaint Management	Collection verification and dispute handling.	ILO-02, ILO-05
4	Fertilizer & Payment Management	Fertilizer workflows, payment calculations, deductions.	ILO-02
5	Reporting, Analytics & Notifications	Dashboards, reports, exports, and notifications.	ILO-02, ILO-03

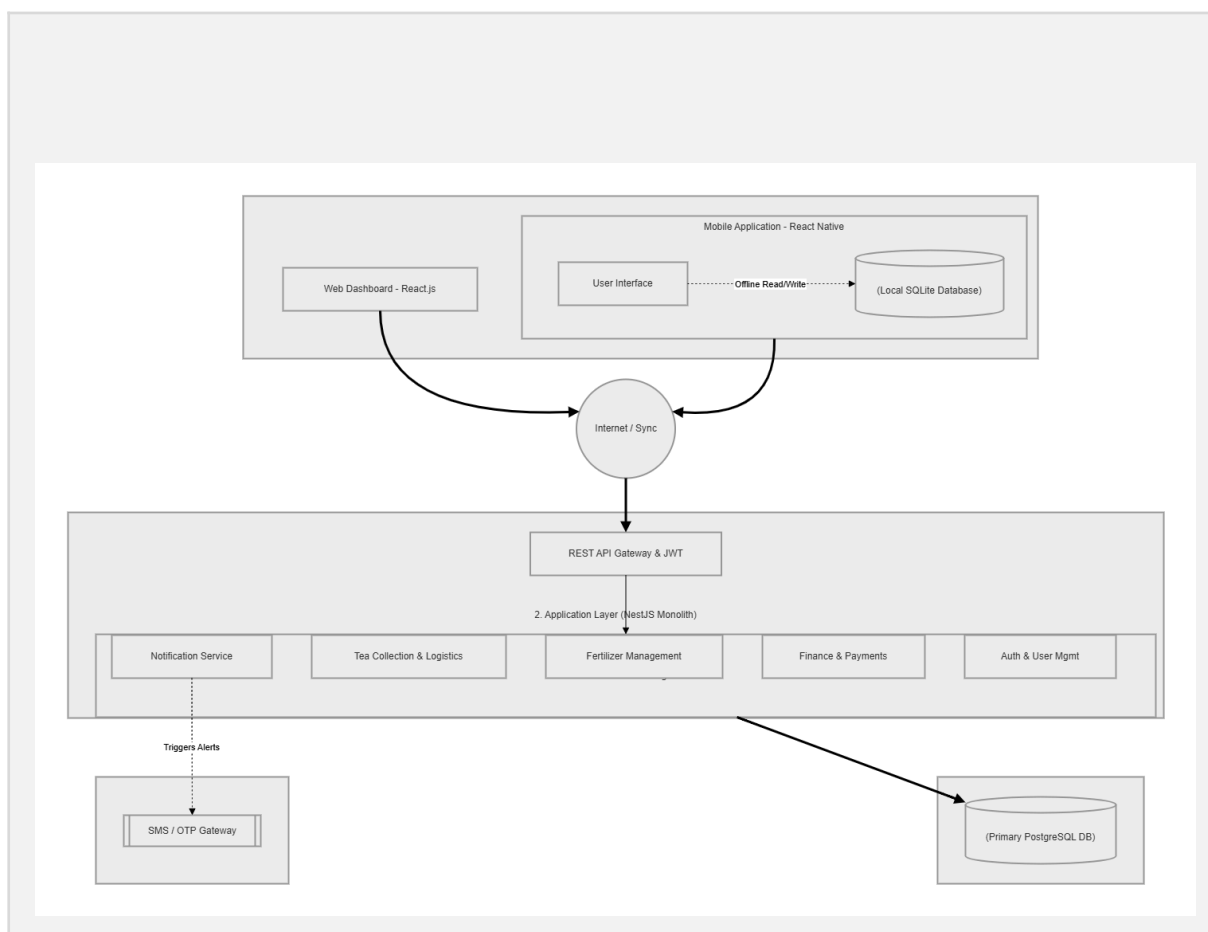
D. TECHNICAL APPROACH

D1. Technology Stack

Layer	Technology / Framework	Justification
Frontend / UI	React Native, TypeScript ,React.js	The system uses a React.js web app for factory administration and a React Native mobile app for field users with offline support via SQLite.
Backend / API	Node js, NestJS, REST API	High-performance, event-driven processing suitable for handling multiple concurrent requests. NestJS offers a modular and scalable architecture with built-in support for dependency injection, security, and maintainability. REST APIs

Layer	Technology / Framework	Justification
		ensure efficient communication between mobile clients and the server.
Database	PostgreSql,SQLite	PostgreSql is used as the central database because it provides reliable, secure, and scalable storage for managing all system data efficiently. SQLite is used for local mobile storage, allowing the application to operate offline and synchronize data with the central database when internet connectivity is available.
DevOps / Deployment	Cloud Hosting, GitHub	Version control, collaboration, code review, and project management capabilities. Cloud hosting ensures system availability, scalability, remote accessibility, and simplified deployment and maintenance.

D2. System Architecture Overview



E. PROJECT PLAN & WORKLOAD DISTRIBUTION

E1. High-Level Timeline (2 Semesters)

Phase	Milestone / Deliverable	Target Week(s)	Responsible Member(s)
1	Requirements Gathering, Stakeholder Analysis, Project Proposal	1 - 4	All members
2	System Analysis and Design	5 - 8	All members
3	Core System Development	9 - 12	All members
4	Tea Collection & Delivery Module Development	13 - 15	All members
5	Fertilizer, Payment & Complaint Module Development	16 - 22	All members
6	Reporting, Notifications & System Testing	23 - 27	All members
7	Deployment, Documentation & Final Presentation	28 - 30	All members

E2. Individual Contribution Plan

Student Name	Primary Responsibilities	Contribution Area	Est. Hours
W. D. R. Migara	UI/UX design, Front end development	☒ SE ☐ BA ☐ Both	≥ 500 hrs
R.E.M.K.R.B Ekanayake	Backend development	☒ SE ☐ BA ☐ Both	≥ 450hrs
G.S.Naarayanan	Business Analysis,project management	☐ SE ☒ BA ☐ Both	≥ 400 hrs
U.H.N.Rodrigo	Backend Development	☒ SE ☐ BA ☐ Both	≥ 450s
S.R.A.Patabedi	DevOps Engineering, Software Design, Backend development	☒ SE ☐ BA ☐ Both	≥ 500 hrs

F. INTENDED LEARNING OUTCOME ALIGNMENT

ILO	Description	Status
ILO-01	Synthesise project requirements into structured execution plans with defined milestones and task distribution.	[✓] Addressed
ILO-02	Build functional, scalable software by applying engineering principles, modern frameworks, and deployment strategies.	[✓] Addressed
ILO-03	Validate software integrity through rigorous testing and maintain industry-standard technical documentation and version control.	[✓] Addressed
ILO-04	Demonstrate professional communication and effective teamwork to achieve project objectives.	[✓] Addressed
ILO-05	Resolve complex technical challenges through research and critical thinking while reflecting on professional growth.	[✓] Addressed

Brief explanation of ILO coverage:

ILO-01: The project includes requirement gathering, project planning, milestone definition, workload distribution, and progress tracking through a structured development timeline.

ILO-02: The Tea Collection Management System will be developed using React Native, NestJS, and PostgreSQL to build a scalable, secure, and functional software solution.

ILO-03: Software quality will be ensured through unit testing, integration testing, system testing, API validation, GitHub version control, and comprehensive project documentation.

ILO-04: Team members will collaborate throughout requirements analysis, development, testing, and presentations while maintaining regular communication with supervisors and industry stakeholders.

ILO-05: The project addresses real-world challenges in tea collection operations by researching domain requirements, designing effective workflows, and implementing solutions for collection tracking, OTP verification, fertilizer management, and payment processing.

G. RISKS, CONSTRAINTS & ASSUMPTIONS

#	Risk / Constraint	Likelihood (H/M/L)	Mitigation Strategy
	Poor internet connectivity in plantation areas	H	Optimize the application for low-bandwidth usage and minimize data transfers.
	User resistance to adopting a digital system	M	Provide a simple user interface and user training sessions.
	Security risks and unauthorized access	L	Use JWT authentication, role-based access control, and secure API practices.

H. TRACK-SPECIFIC DECLARATIONS**H1. UCSC Track (UT)**

UGVLE Topic Reference No.:

Startup under UCSC Track

Mentor Name & Organisation:

Name: Sachith Kasthuriarachchi

Organization: WSO2

H2. Industry Track (IT)

Company Name:

PDC Registration No.:

Agreement Signed? ☐ Yes ☐ No

H3. Startup Track (ST)

Registered Company Name:

PDC Registration No.:

Founding Member(s) in Group? ☐ Yes

I. DECLARATIONS & SIGNATURES

We confirm that the information provided is accurate, that this project is a genuine software development initiative, and that we have read and agree to the Industry Project terms and conditions.

#	Student Name	Signature	Date
1	W. D. R. Migara		03.06.2026
2	G.S.Naarayanan		03.06.2026
3	U.H.N.Rodrigo		03.06.2026
4	R.E.M.K.R.B Ekanayake		03.06.2026
5	S.R.A.Patabedi		03.06.2026

Academic Co-Supervisor Name:

Signature & Date:



03/6/2026

Industry Mentor / Supervisor Name: Sachith
Nilmantha Kasthuriarachchi



Signature & Date: 03.06.2026